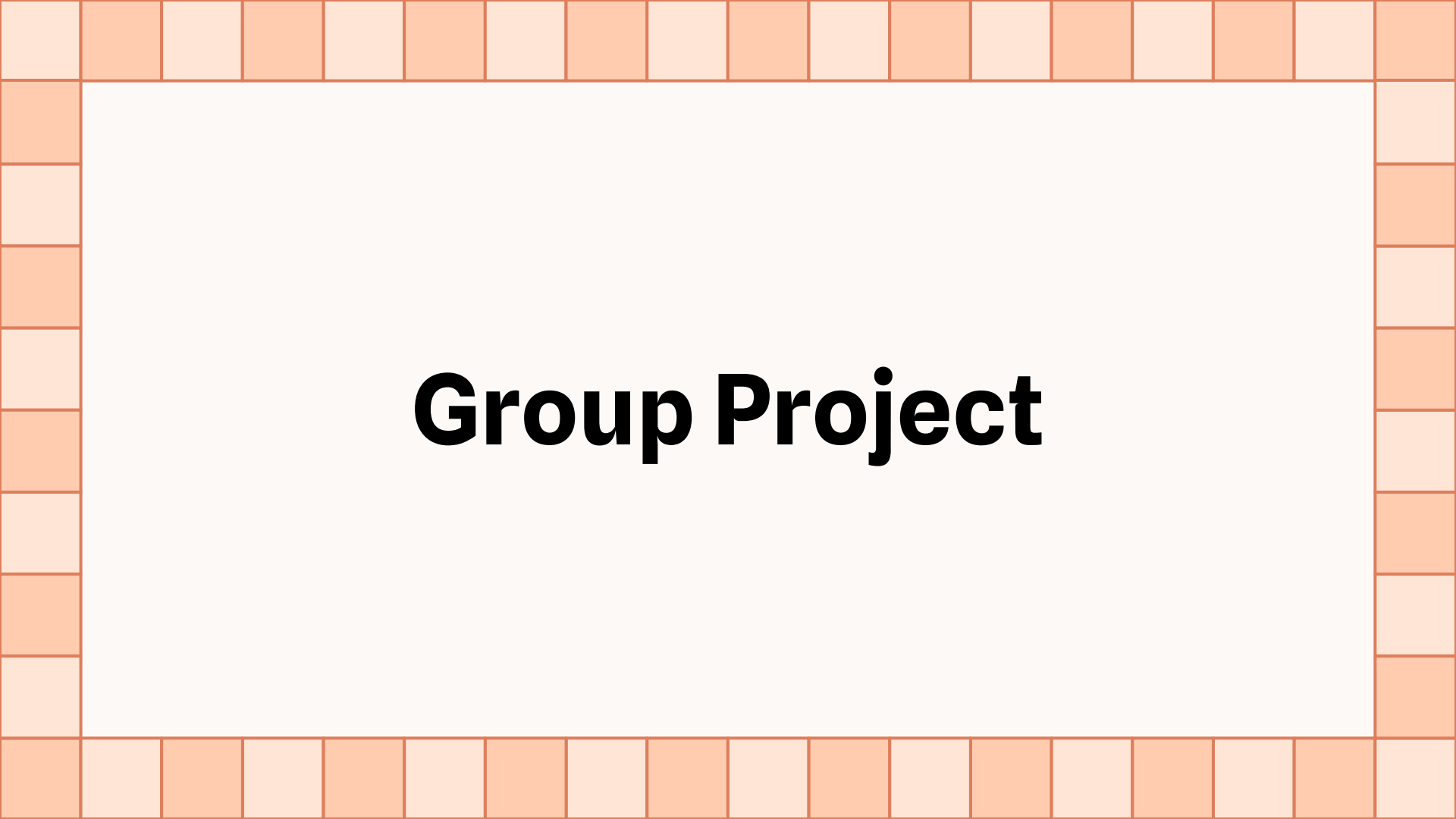




Data and Datasets

A decorative border composed of a grid of squares in two shades of orange, surrounding a central white area.

Group Project

What to expect

Last week, we read seven chapters from *Computational Humanities* (2024) and talked about various DH projects, introducing you to the variety of methods and approaches in this field.

I invited you to think about what kinds of DH projects interest you.

Our goal today is to talk about data and datasets. You can't do a DH project without a dataset.

After this class meeting, I will create Slack channel, where you will post a sentence or two about the dataset(s) you are interested in working with and some initial questions for your group project. Your task is to read each others' posts and form your groups.

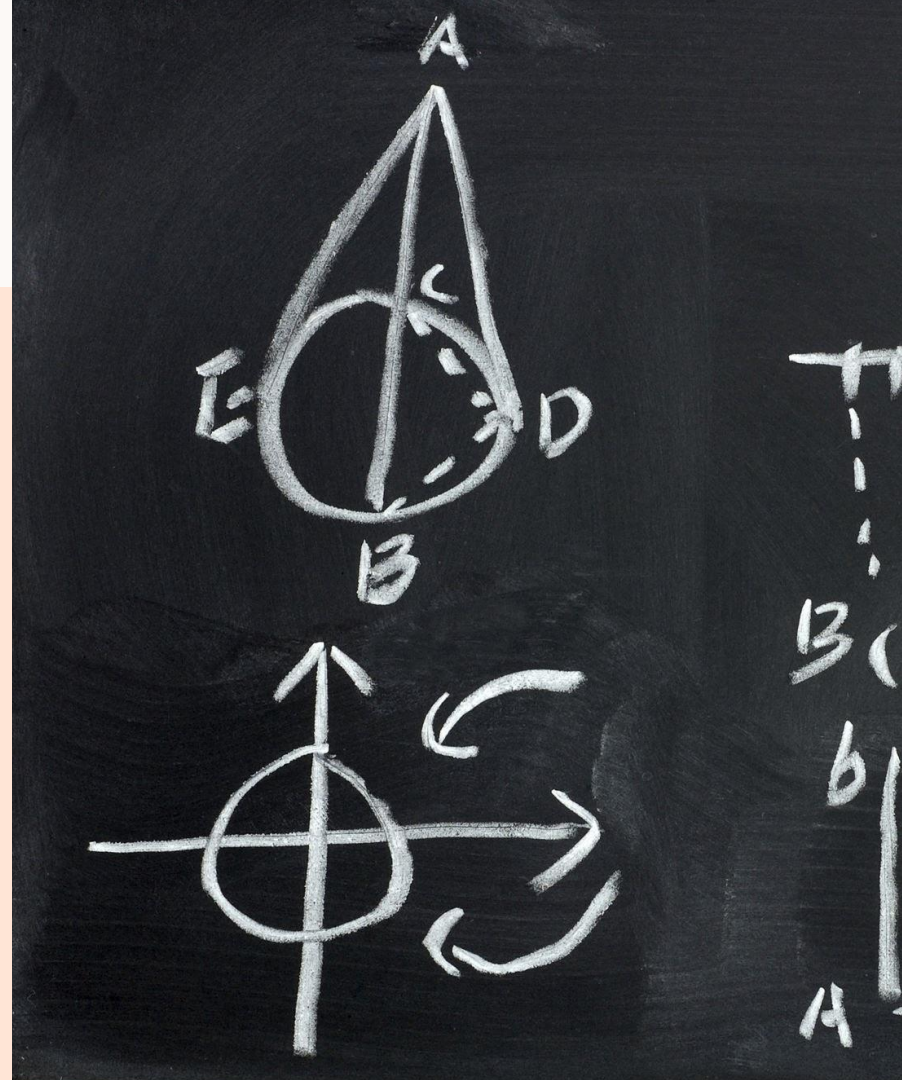
Next Assignment

Group Project - Topic Proposal Due: Friday, October 9

This assignment is **two-pages long**, ca. 500 words.

It should contain the following:

- Names of the group members (incl. their names as it appears on CourseLink)
- Dataset A link to the dataset that you will be using in this assignment and a description of the dataset and why you chose it
- Initial ideas What are some of the first questions that come to your mind? What directions will you explore? Where will you begin your analysis?



From the Course Outline

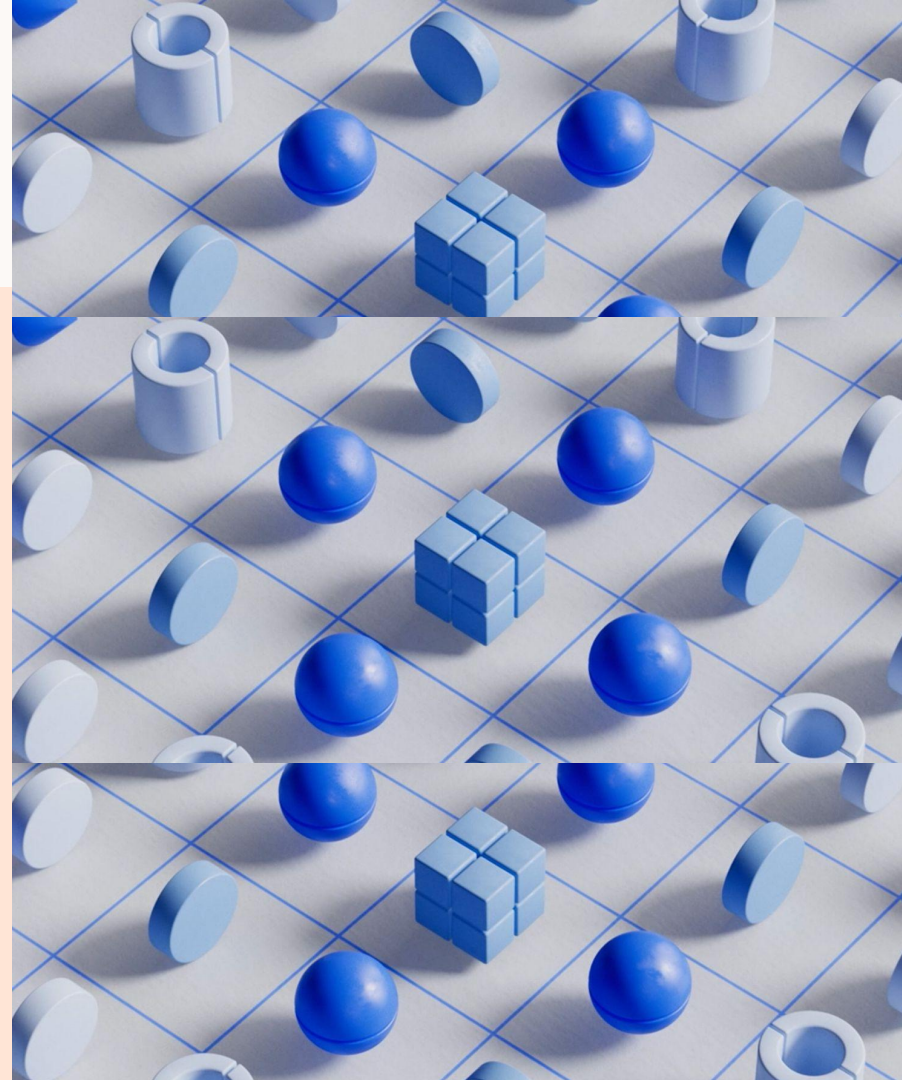
Topic statement is the first assignment for the group project. It should describe the motivating research idea, include your initial research question(s) and identify the dataset. Please offer a well-grounded reasoning on why this dataset is relevant for this research direction. We will discuss how to evaluate datasets in class. **You must identify all the members of your group beforehand and devise a plan for the division of labor (not included in the page count). Division of labor should not mean one student doing the writing and another doing the coding. You must split the tasks equitably.** Upon forming your groups, you will be put into a Slack channel with your group members. The topic statement should be 2 pages long.

Definitions

What is Data?

The term 'data' comes from the plural of 'datum', a Latin word meaning 'something given.'

“Data is a **collection of facts, numbers, words, observations or other useful information**. Through data processing and data analysis, organizations transform raw data points into valuable insights”
(IBM 2024,
<https://www.ibm.com/think/topics/data>)

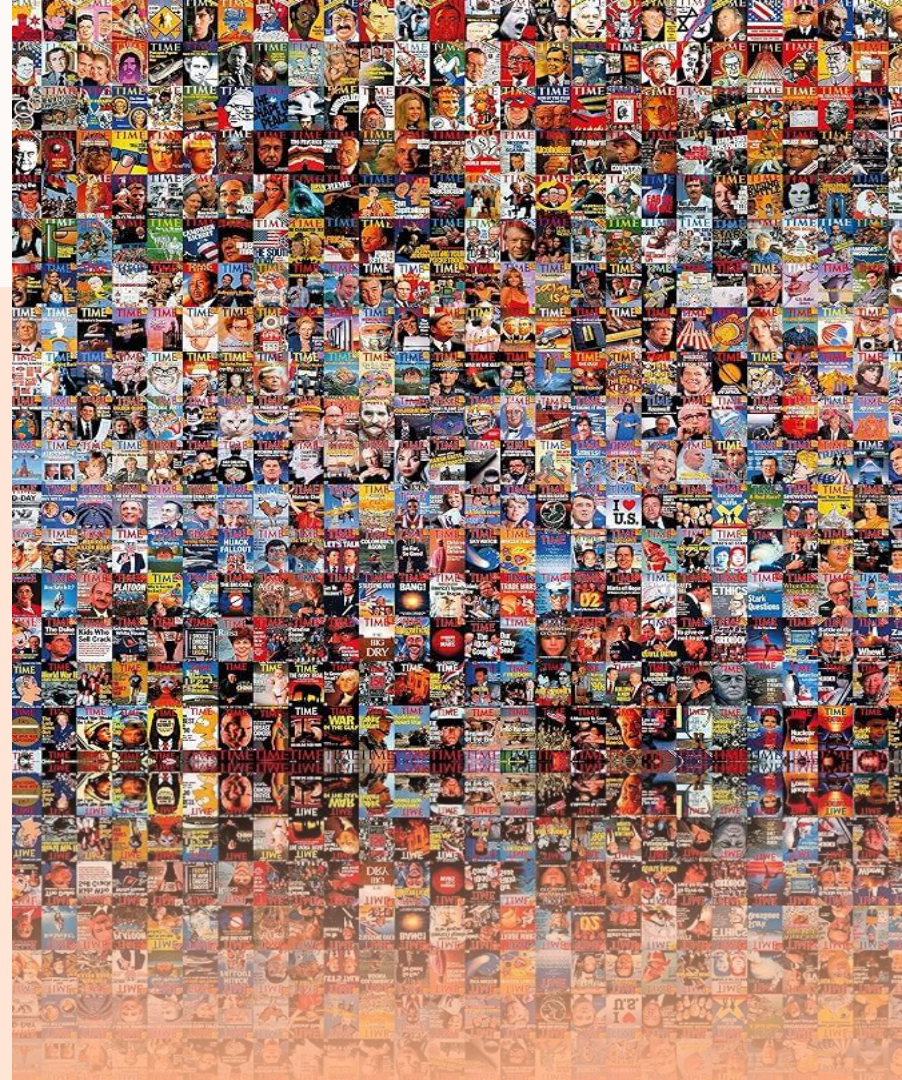


Data as Capta

“*Capta* is ‘taken’ actively while *data* is assumed to be a ‘given’ able to be recorded and observed.”

“Rendering *observation* (the act of creating a statistical, empirical, or subjective account or image) as if it were *the same as the phenomena observed* collapses the critical distance between the phenomenal world and its interpretation, undoing the basis of interpretation on which humanistic knowledge production is based.”

Johanna Drucker, “Humanities Approaches to Graphical Display” DHQ (2011)



Data in DH

Data Feminism

“a way of thinking about data, both their uses and their limits, that is informed by direct experience, by a commitment to action, and by intersectional feminist thought” - D’Ignazio and Klein (2020)

Collections as data

“A collections as data paradigm seeks to foster an expanded set of research, pedagogical, and artistic potential predicated on the computational use of cultural heritage collections.” - Padilla (2018)

FAIR Principles: Findability, Accessibility, Interoperability, and Reusability

Supporting the reuse of scholarly data. - Wilkinson, et al. (2016)

CARE Principles for Indigenous Data Governance: Collective Benefit, Authority to Control, Responsibility, and Ethics

“integrate Indigenous knowledges and approaches into data practices and policies” - Carroll, et al. (2020)

Links for the readings relevant to the previous slides

- <https://data-feminism.mitpress.mit.edu/>
- <https://dhq.digitalhumanities.org/vol/5/1/000091/000091.html>
- <https://www.ibm.com/think/topics/data>
- <https://crln.acrl.org/index.php/crlnews/article/view/17003>
- <https://www.nature.com/articles/sdata201618>
- <https://www.nlm.nih.gov/oet/ed/cde/tutorial/02-200.html>
- <https://datascience.codata.org/articles/10.5334/dsj-2020-043>
- <https://journalofdigitalhumanities.org/2-3/big-smart-clean-messy-data-in-the-humanities/>
- <https://miriamposner.com/blog/humanities-data-a-necessary-contradiction/>

18th century origin of statistics as “Staatswissenschaft”

Gottfried Achenwall argued that *Staatswissenschaft* (1749) or *Statistik* (1751), is about learning useful insights about a state from countless matters that one encounters in a state body. His method was quantification. Achenwall emphasized the importance of counting changes within the territories of a state: population trends, like births and deaths, administrative units, like villages, towns, and cities, and production and trade of goods and products. Quantification acted as an abstraction and allowed these otherwise distinct aspects of the state body to be understood and studied on a shared basis.

Let's complicate '*data*'

Input: Artifact

An artifact is something that has been **created, shaped, or modified** by human beings and holds significance or meaning.

This line of argumentation could be expanded to natural phenomenon, events, and observations more generally but they are not our topic today.

Process: Quantification

Quantification is the process of transforming these artifacts into **measurable, categorized** units. It allows **numerical questions** and enables the analysis of these artifacts through a **unified, quantitative** framework.

We can think of letters as texts to be read and enjoyed or as collections of words to be counted.

Output: Data

Not everything becomes data and not everything that becomes data is valued in the same way.

Creating data involves **decisions** about what to count, measure, and categorize, shaping the data based on **societal and academic norms**.

A **dataset** refers to a structured collection of **quantified and categorized data**. Datasets contain data in various forms. They come in all shapes and sizes. Not all datasets have to be '*big*.' What matters is that your analysis and conclusions are in **proportion** to the kind of dataset that you studied.

Metadata refers to the **descriptive information about the dataset** itself, rather than the actual data it contains.

Metadata provides contextual details that explain **how, where, and why the data was collected**, as well as the structure of the dataset.

If the dataset includes any **annotations** (like themes, genre, period, etc.), metadata should describe **how the data was categorized or labeled**. It should also include the provenance or **the history of the data**, including any alterations or updates.

A decorative border composed of a grid of squares in two shades of orange, surrounding a central white area.

Choosing a dataset

What to look for in a dataset?

A

What is in this dataset? Does it contain text files? Images? Spreadsheets? How many data points? What is/are the language(s) of the files? When were they created?

B

Who created this dataset and why? Are they clear about the decisions that went into the process of making this dataset?

C

How do you access this dataset? What are various rules and standards that govern the use of this dataset? How is this dataset shared?

D

What are some of use cases? What kind of research did the authors who published this dataset conducted with it?

Example: UNESCO's Proceedings, 1945–2017: A Bilingual Digital Text Corpus

The screenshot shows the article page for 'UNESCO's Proceedings, 1945–2017: A Bilingual Digital Text Corpus' in the Journal of Open Humanities Data. The page features a header with the journal title, submission and review buttons, and the Juhlubiquity press logo. Below the header, the article title is displayed, followed by the authors' names: Benjamin G. Martin, Fredrik Mohammedi Norén, Roger Mähler, and Andreas Marklund, Oriane Martin. The abstract section provides a detailed description of the digital text corpus, highlighting its metadata, machine-readability, and storage on Zenodo and GitHub. The keywords section lists 'digital text analysis', 'international organizations', 'transnational history', 'global humanities', 'text corpus', and 'corpus design'. The footer includes the year 2025, volume 11, page/article 31, and DOI 10.5334/johd.314. It also shows the submission date (Feb 5, 2025), acceptance date (Mar 26, 2025), and publication date (May 5, 2025). The page is marked as 'Peer Reviewed' and includes a 'CC Attribution 4.0' license icon.

Journal of Open Humanities Data Start Submission Become a Reviewer Juhlubiquity press

Reading: UNESCO's Proceedings, 1945–2017: A Bilingual Digital Text Corpus Download A+ A- 99 Alt Display

Data Paper
UNESCO's Proceedings, 1945–2017:
A Bilingual Digital Text Corpus

Benjamin G. Martin, Fredrik Mohammedi Norén, Roger Mähler,
Andreas Marklund, Oriane Martin

Abstract
The record of the meetings of UNESCO's General Conference offers a valuable resource for research in the global humanities. We present a digital text corpus, including metadata and supplementary material, that makes the complete record of these meetings from 1946 to 2017 in English and/or French accessible in a machine-readable form that is suitable for digital text analysis. The corpus is stored on Zenodo; relevant code is available on GitHub. The corpus offers reuse potential for scholars interested in any of the countless issues that have been discussed and debated in UNESCO's General Conference over more than seventy years, as well as to Natural Language Processing (NLP) developers interested in the challenges of language recognition and automated segmentation.

Keywords: digital text analysis, international organizations, transnational history, global humanities, text corpus, corpus design

Year: 2025 Volume 11 Page/Article: 31 DOI: 10.5334/johd.314

Submitted on Feb 5, 2025 Accepted on Mar 26, 2025 Published on May 5, 2025

Peer Reviewed CC Attribution 4.0

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<https://openhumanitiesdata.metajnl.com/articles/10.5334/johd.314>

<https://zenodo.org/records/147866>
89

GitHub Repository

Follow the link from
the dataset paper:
https://github.com/inidun/proceedings_curation/releases

The screenshot shows the GitHub repository page for 'inidun / proceedings_curation'. The repository is at version v0.2.0. The main section displays 'Proceedings Curation v1.0' as the latest release, with a 'Compare' button. Below this, the 'Assets' section lists several files: 'meetings_eng.zip' (30.2 MB, Nov 27, 2024), 'meetings_eng_fre.zip' (47.4 MB, Dec 10, 2024), 'meetings_fre.zip' (16.6 MB, Nov 27, 2024), and 'metadata_index.csv' (986 KB, Dec 10, 2024). The description of the release mentions that tokens were counted using 'nltk.tokenize.word_tokenize'.

inidun / proceedings_curation

Type to search

<> Code Issues Pull requests Actions Projects Security Insights

Releases / v0.2.0

Proceedings Curation v1.0 Latest Compare

albakeneko released this Nov 27, 2024 · 4 commits to main since this release v0.2.0 1b80c21

Assets

- meetings_eng.zip : Corpus consisting of English text extracted from UNESCO's Proceedings, 1945-2017. Contains 15.7M tokens. Uncompressed size: 83.9 MB.
- meetings_fre.zip : Corpus consisting of French text extracted from UNESCO's Proceedings, 1945-2017. Contains 8.6M tokens. Uncompressed size: 48.3 MB.
- meetings_eng_fre.zip : Corpus consisting of English and French text extracted from UNESCO's Proceedings, 1945-2017
- Proceedings OCR Quality Control_Overview.xlsx : An overview of the OCR quality of the Proceedings PDFs
- metadata_index.xlsx : An overview of all sessions and meetings (XLSX format)
- metadata_index.csv : An overview of all sessions and meetings (CSV format)

Tokens were counted using `nltk.tokenize.word_tokenize`.

Assets 8

meetings_eng.zip	30.2 MB	Nov 27, 2024
meetings_eng_fre.zip	47.4 MB	Dec 10, 2024
meetings_fre.zip	16.6 MB	Nov 27, 2024
metadata_index.csv	986 KB	Dec 10, 2024

UNESCO Proceedings

It is crucial to understand where
the data is coming from

Records of the General Conference of
UNESCO, second session, Mexico, 1947,
v. 1: Proceedings,

<https://unesdoc.unesco.org/ark:/48223/pf0000214793.locale=en>

The CHAIRMAN : You have heard the motion, ladies and gentlemen. Any discussion ? Then we shall take the vote. All in favour kindly hold up their hands. Against ? The motion is *carried*.⁽¹⁾

M. MARITAIN (France) (*Translation from the French*) : I shall read the second part of Resolution IV :

(2) that Round Table Discussions dealing with topics proposed in the final resolution of the Round Table Discussions in Mexico be organized by Unesco in 1948, in conjunction with the Amsterdam International Congress of Philosophy;

(3) that Unesco make arrangements with a number of philosophical reviews, or general cultural reviews, of a high order and of international circulation, so that these may devote special issues to symposia dealing with these topics. A suitable number of copies of such issues would be distributed by Unesco in connexion with the Discussions referred to in No. 2.

The CHAIRMAN : The motion is seconded by the delegate of the United States. It is a little difficult for the meeting to discuss these matters. For instance, No. 2, Round Table Discussions, deals with « topics » but we do not know what those topics are.

These resolutions, i. e. Nos. 2 and 3 of Resolution IV, are now open for discussion. If there is no discussion, will all those in favour of the resolution kindly hold up their hands. Resolutions 2 and 3 are carried, consequently Resolution IV as a whole is *carried*.⁽²⁾

M. MARITAIN (France) (*Translation from the French*) : Mr. Chairman, with your permission we can now turn to Resolution V, reading :

The Working Group on Translations, in the light of the Report on Translations (Doc. 2C/41) and its deliberations thereon, resolves as follows :

(1) In view of the importance of translations of classics of various countries for international understanding, Unesco will in 1948 put into effect the measures set forth in document 2C/41 for the translation of classics.

(2) The Director-General is therefore instructed in accordance with the resolution of the Economic and Social Council :

(a) in the light of the work of the Conference of Allied Ministers of Education, to take measures necessary for the preparation of a co-ordinated plan for the translation of classics, for submission to the Economic and Social Council by 1 June 1948, and keep the Council informed of progress of work on this project, seeking in turn all possible aid for the project from the United Nations Organization ;

(b) to prepare plans for ensuring adequate translation of :
(i) the great works of classic reputation to which the Economic and Social Council refers ;

(ii) contemporary works not only in the field of literature, but also in those of philosophy and of social and natural science, the diffusion of which is of extreme importance for international understanding ;

⁽¹⁾ In final form : Resolutions 2.3.3.4. : vol. II, page 16 and 4.3.2.2. : page 21.

⁽²⁾ In final form : Resolutions 4.3.1.1. : vol. II, page 23 and 4.3.1.2. : page 24.

(c) to summon an International Committee of Experts to review recommendations from Member States and other sources, and prepare the final list and proposals and to report ;
(d) To examine and seek means by which to assist smaller countries in their participation in this project.

(3) The General Conference recommends to Member States that, through their National Commissions, they set up each a Committee of Experts to prepare lists of literary, philosophic and scientific works with a view to eventual translation, and co-operate with Unesco through the various stages of the project.

The CHAIRMAN : Does anyone second this ? It is seconded by the delegate of the United States.

Does anyone wish to speak to it ? The delegate of Canada.

Mr. S. D. PIERCE (Canada) : I would like to make the same point that has been made before, where mention is made of National Commissions. The point is that not all countries have National Commissions, and therefore paragraph 3 should be amended to cover the case of countries which have no National Commissions.

It might be amended by saying « through their National Commissions where they exist », or using the words which we used on another occasion, « that, where possible, through their National Commissions. . . »

The CHAIRMAN : I take it the mover of the motion would have no objection to the words « where possible » being introduced into the second line of paragraph 3 ?

Is there anyone now who wishes to speak to this Resolution V ?

If not, I will put it to the vote. All in favour ? Against ?

The resolution is *passed*.⁽³⁾

While we are on this particular point I must now again give the opportunity to anyone else who wants to raise a matter concerned with Translation.

If there is none, we have finished the whole of that section on Translation in Chapter I.I.D.1. There is still « Languages » in I.I.D.1., which has not yet been touched. The delegate of France.

M. MARITAIN (France) (*Translation from the French*) : Mr. Chairman, there is no other formal proposal arising from the discussions of the Working Party to submit as a resolution.

I only want to remark that there is a general motion, in the seventh part, page 11 of the English text, which approves as a whole a number of basic documents submitted by the Secretariat. I do not think, however, this need be submitted to the present Commission, since it was elaborated and drafted

⁽³⁾ In final form : Resolution 4.2. : vol. II, page 23.

What to look for in a dataset?

A

What is in this dataset? Does it contain text files? Images? Spreadsheets? How many data points? What is/are the language(s) of the files? When were they created?

B

Who created this dataset and why? Are they clear about the decisions that went into the process of making this dataset?

C

How do you access this dataset? What are various rules and standards that govern the use of this dataset? How is this dataset shared?

D

What are some of use cases? What kind of research did the authors who published this dataset conducted with it?

What is in this dataset?

A

This dataset contains **text files** in the form of **a bilingual digital text corpus**. The corpus includes the verbatim records from **UNESCO's General Conference Proceedings**, from 1946 to 2017, in **English and French**. The dataset has two corpora: one in English (approximately **15.7 million tokens**) and one in French (approximately **8.6 million tokens**), totaling around 21 million words. The corpus was compiled between February 2021 and December 2024.

Who created this dataset? Why? How?



B

The dataset was created by researchers in several Swedish universities, with Benjamin G. Martin from **Uppsala University** as the PI. The creation of the dataset was part of the **INIDUN project (International Ideas at UNESCO)**, which aimed to make UNESCO's General Conference Proceedings accessible for digital text analysis. The authors are clear about their decisions in the dataset creation process. **The OCR reprocessing and quality control steps are explained, detailing the procedures for checking OCR accuracy and improving the metadata.** They also describe the extraction process, the design of **scripts** for text extraction by language, and the approach for curating metadata.

How do you access this dataset?

The logo consists of four light orange squares in a row. The third square from the left is a darker shade of orange and contains a white capital letter 'C'.

The dataset is publicly available and can be accessed through the following links:

Zenodo: <https://zenodo.org/records/14786689>

GitHub repository: https://github.com/inidun/proceedings_curation/releases

The dataset is shared under the **Creative Commons Attribution 4.0 International (CC BY 4.0) license**, which allows users to **freely use, share, and adapt the data, as long as proper attribution is given to the creators**. There are no specific restrictions or rules mentioned beyond the attribution requirement, but users are encouraged to follow the project's code and documentation available on GitHub.

What are some use cases?



D

The dataset has **high reuse potential** for digital humanities and social sciences. It supports text analysis (e.g., word-trend analysis, topic modeling, word embeddings) to **study global debates at UNESCO's General Conference**. It also aids historical and transnational research on UNESCO's role in global discussions about **literacy, heritage, and communications**. Additionally, the corpus provides a rich resource for developing **Natural Language Processing (NLP) methods**, such as language recognition, segmentation, and OCR error correction. The authors used the dataset for research on **global conceptual history and international ideas at UNESCO**, enhancing interdisciplinary studies in global humanities.

Where to find more datasets?

- <http://dhresourcesforprojectbuilding.pbworks.com/w/page/69244469/Data%20Collections%20and%20Datasets>
- <https://introdigitalstudies2024.commons.msu.edu/datasets/>
- <https://zenodo.org/>
- <https://www.hathitrust.org/> or <https://analytics.hathitrust.org/>
- <https://www.gutenberg.org/> - books in plain text
- <https://borealisdata.ca/dataverse/guelph>
- <https://jimclifford.ca/hgiscanada/>
- <https://www.lib.uoguelph.ca/find/data-maps/> & <https://guelphhumber.libguides.com/az/databases>



**Find your groups on
Slack!**